Lecture 6

Sub-gradient optimization for ML

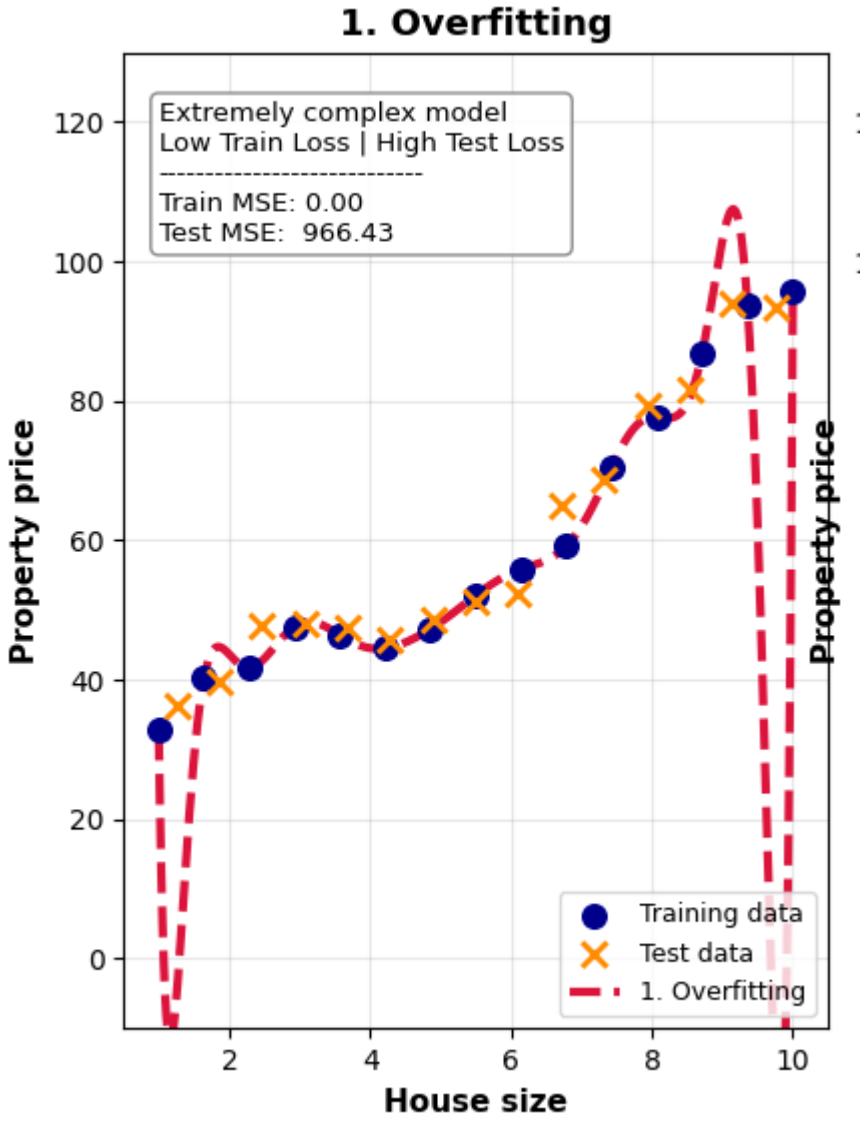
Case study:

- Lasso loss function
- SVM
- Hindge loss function
- Relu Activation function in NN.

Sub-gradient and l_1 regularization

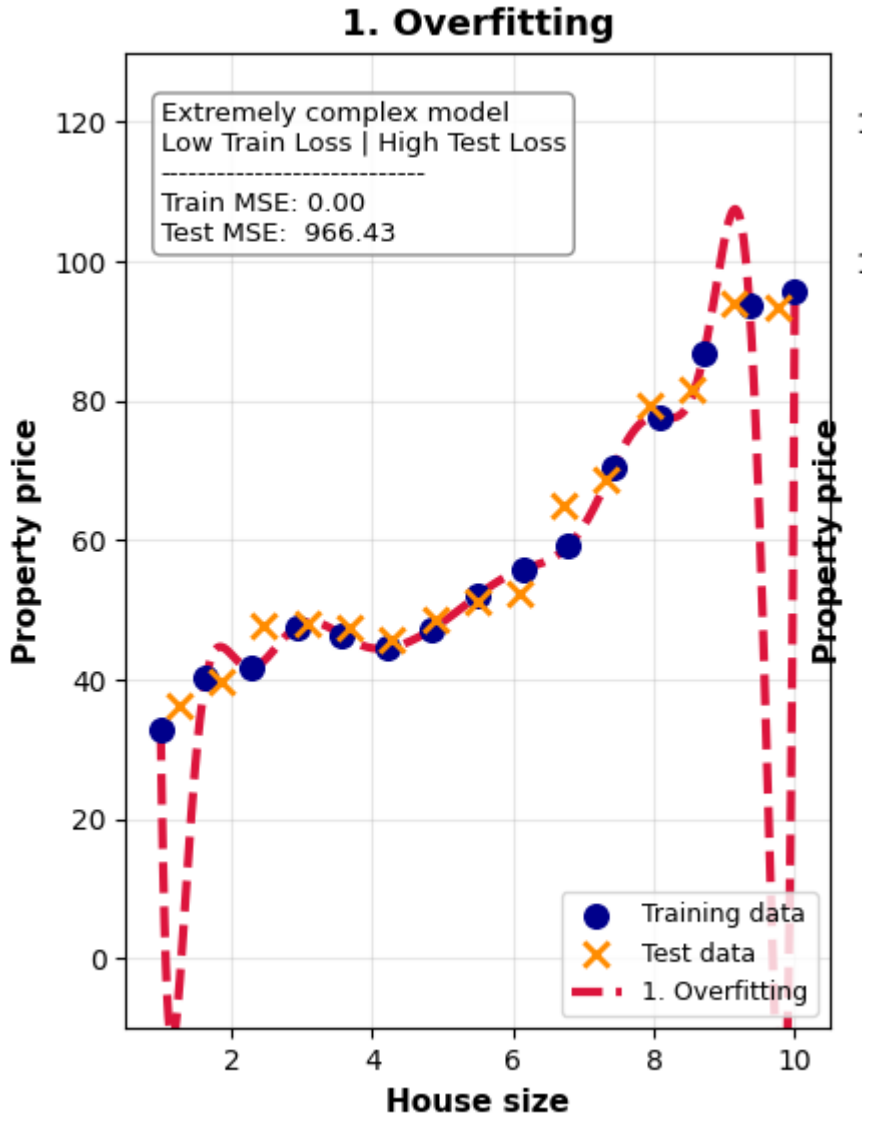
- **Sub-gradient optimization algorithm:**
- A subgradient is an algorithm for minimizing a **non-differentiable convex function**.
- A subgradient is a **generalization of the gradient** for non-differentiable **convex functions**. If a function is convex but not differentiable at a point, a subgradient can be used instead.

- **Regularization:**
- Regularization is a technique in machine learning and optimization that introduces additional constraints or penalties into the model's **training process** to **prevent overfitting, improve generalization** and **reduce variance**. By discouraging the model from fitting the noise in the training data, regularization helps the model to generalize better to **unseen data**.



Regularization

- For a general loss function $L(\widetilde{W})$ (mean square, cross entropy, etc.), where $\widetilde{W}$ represents the model's parameters.
- Under l_1 **regularization (Lasso),** the regularized loss function is
  - $\underset{\widetilde{W}}{arg\,min}\{L(\widetilde{W})\}$ **subject to** $\|\widetilde{W}\|_1 \leq t$
  - **Or** $L_{l_1}(\widetilde{W}) = \underset{\widetilde{W}}{arg\,min}\left\{L(\widetilde{W}) + \lambda\|\widetilde{W}\|_1\right\}$
- **Under** l_2 **regularization (Ridge)**, the regularized loss function is
  - $\underset{\widetilde{W}}{arg\,min}\{L(\widetilde{W})\}$ **subject to** $\|\widetilde{W}\|_2 \leq t$
  - **Or** $L_{l_2}(\widetilde{W}) = \underset{\widetilde{W}}{arg\,min}\left\{L(\widetilde{W}) + \lambda\|\widetilde{W}\|_2\right\}$
- Large regularization coefficient λ: Forces parameters $\widetilde{W}$ close to zero, stripping the model of its ability to learn patterns. Leads to Underfitting.
- Too Little or No Regularization coefficient λ: Allows parameters to grow unchecked, letting the model memorize every random data point/noise. Leads to Overfitting



***Lasso: least absolute shrinkage and selection operator**

If $a = (a_1, a_2, a_3, \dots, a_n) \in \mathbb{R}^n$
l_1 norm: $\|a\|_1 = |a_1| + |a_2| + \cdots + |a_n|$
l_2 norm: $\|a\|_2 = \sqrt{(a_1^2 + a_2^2 + \cdots + a_n^2)} = \sqrt{a^T a}$

Regularization and model fitting:

Model State	Errors	Cause	How Regularization Handles It	Other Solutions
Overfitting	• Low Training Error • High Test Error	Model is too complex, noisy, or flexible (e.g., high-degree polynomial).	**Apply or increase regularization:** Use a larger λ with L1 regularization or L1 regularization to penalize large weights.	• Add more training data • Feature selection • Early stopping
Good Fit	• Low Training Error • Low Test Error	Model complexity appropriately matches the underlying pattern.	**Maintain balance:** Keep the regularization parameter λ at an appropriate value, typically selected using cross-validation.	• Keep the current architecture • Monitor validation/test performance
Underfitting	• High Training Error • High Test Error	Model is too simple, or the regularization penalty λ is too high.	**Reduce regularization:** Decrease λ toward zero or remove the penalty if it excessively restricts model capacity.	• Increase model complexity • Add better features • Use a more powerful algorithm

Convexity: revision

First order Convexity:

A function $f: \mathbb{R}^n \to \mathbb{R}$ is convex if its domain is a **convex set** and for all x_0, x_1 in **its domain**, and all $\lambda \in [0,1]$, we have

$$f(\lambda x_0 + (1-\lambda)x_1) \leq \lambda f(x_0) + (1-\lambda) f(x_1)$$

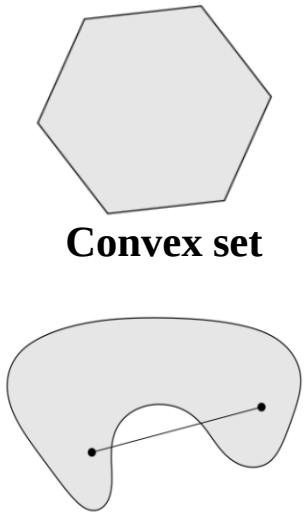
- **Convex set:** A set C is convex if the line segment between any two points in C lies in C, i.e. $\forall x_1, x_2 \in C, \forall \lambda \in [0,1]$

$$\lambda x_1 + (1-\lambda)x_2 \in C$$

First order Convexity:

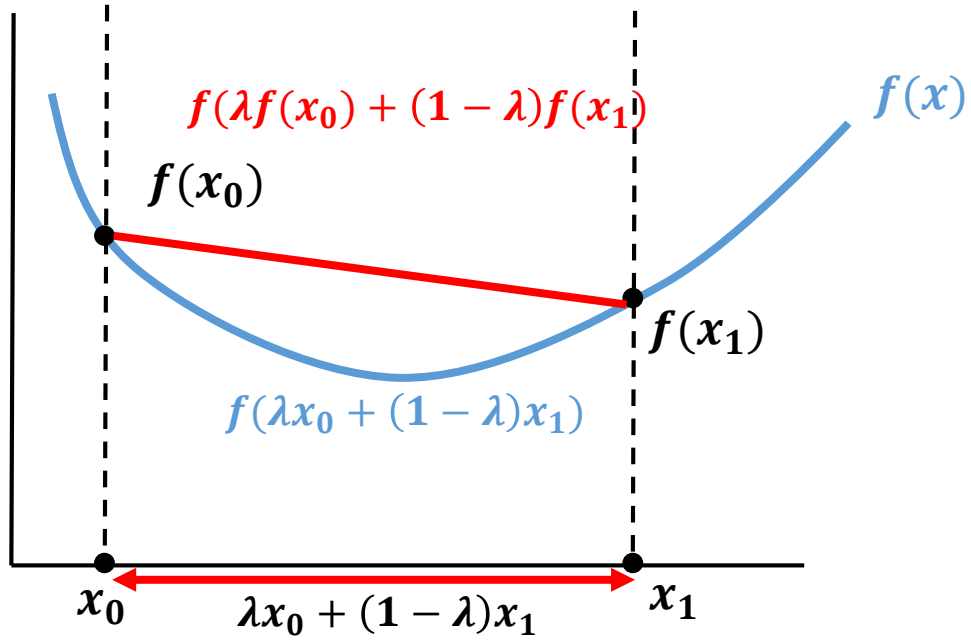
$$f(x + \alpha d) \geq f(x) + \alpha \nabla f(x)^T d, \ \forall x, d \in \mathbb{R}^n, \alpha > 0$$

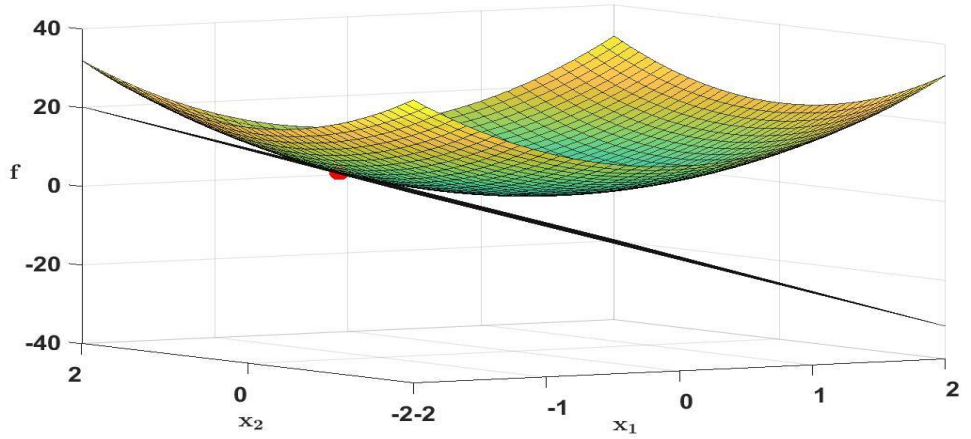
- The function $f(x)$ lies above its **linear approximation** at any point x in $\mathbb{R}^n$.
- **At a local minima x^*, $\nabla f(x^*) = 0$ it means $f(x^* + \alpha d) \geq f(x^*)\ \forall x, d \in \mathbb{R}^n, \alpha > 0 \Rightarrow x^*$ is the global minima**



Convex set

Non-convex set





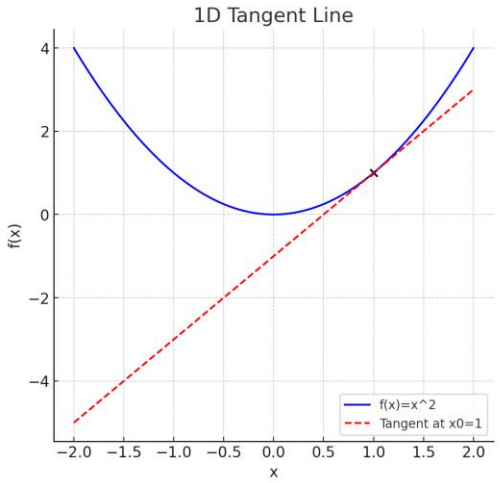
$f(x) = f(x_1, x_2) = 5x_1^2 + 3x_2^2, H = \begin{bmatrix} 10 & 0 \\ 0 & 6 \end{bmatrix}$, *eig=10,6*

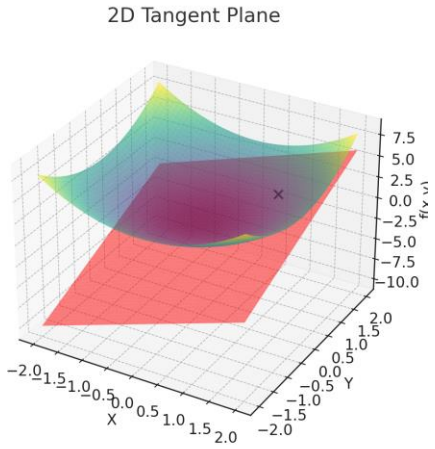
Sub-gradient

Tangent Hyperplane:
For a differentiable function $f: \mathbb{R}^n \to \mathbb{R}$, the tangent hyperplane at a point x_0 is the linear approximation of the function at that point:

$$h(x) = f(x_0) + \nabla f(x)^T (x - x_0)$$

- At a point x_0, the tangent hyperplane is the linear approximation of f around x_0.
- Touches the function at x_0.
- Lies below the function if f is **convex**





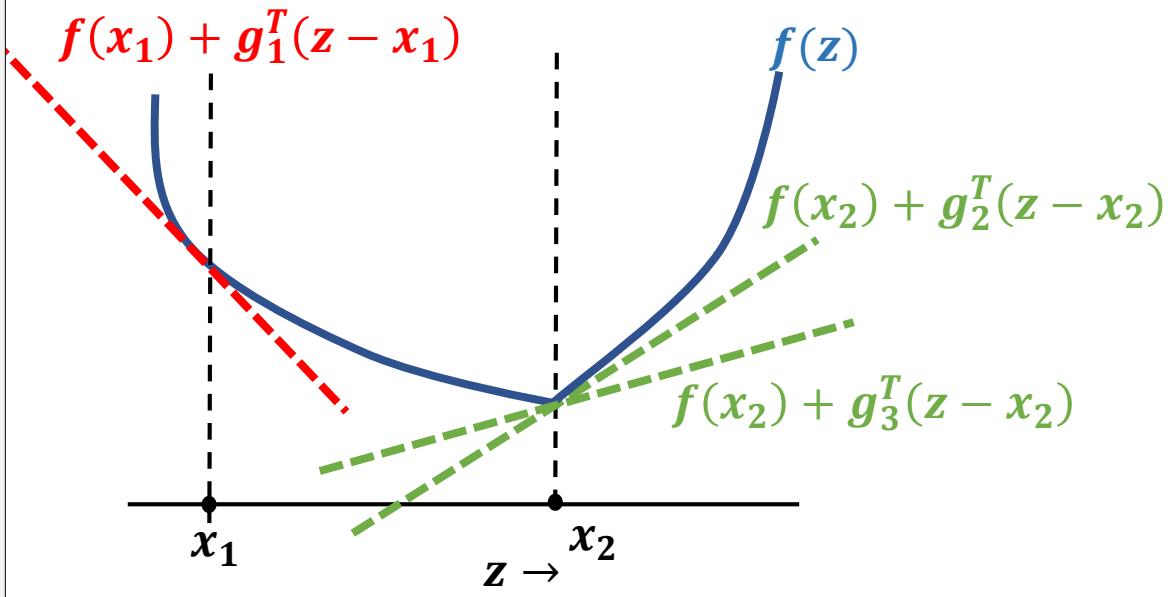
- **First order Convexity:**
  - $f(x + \alpha d) \geq f(x) + \alpha \nabla f(x)^T d, \ \forall x, d \in \mathbb{R}^n, \alpha > 0$
- *It means the function lies above its tangent hyperplane (linear approximation) at every point x.*
- Let, $z = x + \alpha d$, then $d = \frac{(z-x)}{\alpha}$, now first order convexity condition becomes:
  - $f(z) \geq f(x) + \nabla f(x)^T (z - x)$, at $x \in \text{dom } f$ if $\forall z \in \text{dom } f$
  - $f(z) \geq h(z), \forall z \in \text{dom } f$
- *Geometrically: at any x, the **tangent hyperplane** lies below the function everywhere in its domain.*
- *Or the tangent plane is a global underestimator of f.*

Sub-gradient

- **Supporting Hyperplane:**
- A supporting hyperplane generalizes the tangent hyperplane to **convex functions**, **which may not be differentiable everywhere**.
- For a **convex function** f, a hyperplane is supporting at x_0 if:

$$f(z) \geq f(x) + g^T(z - x), \forall z \text{ dom } f$$

  - where g is a **sub-gradient** at x_0
  - The hyperplane "supports" the graph of f from below, the function never goes below this plane.
  - For differentiable convex functions: The tangent hyperplane is also a supporting hyperplane.



Sub-gradient:

- Sub-gradients extend the idea of gradients **to non-differentiable convex functions.**
- A vector $g \in \mathbb{R}^n$ is said to be a sub-gradient of a convex function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ at $x \in \text{dom } f$ if,

$$f(z) \geq f(x) + g^T(z - x), \forall z \in \text{dom } f$$

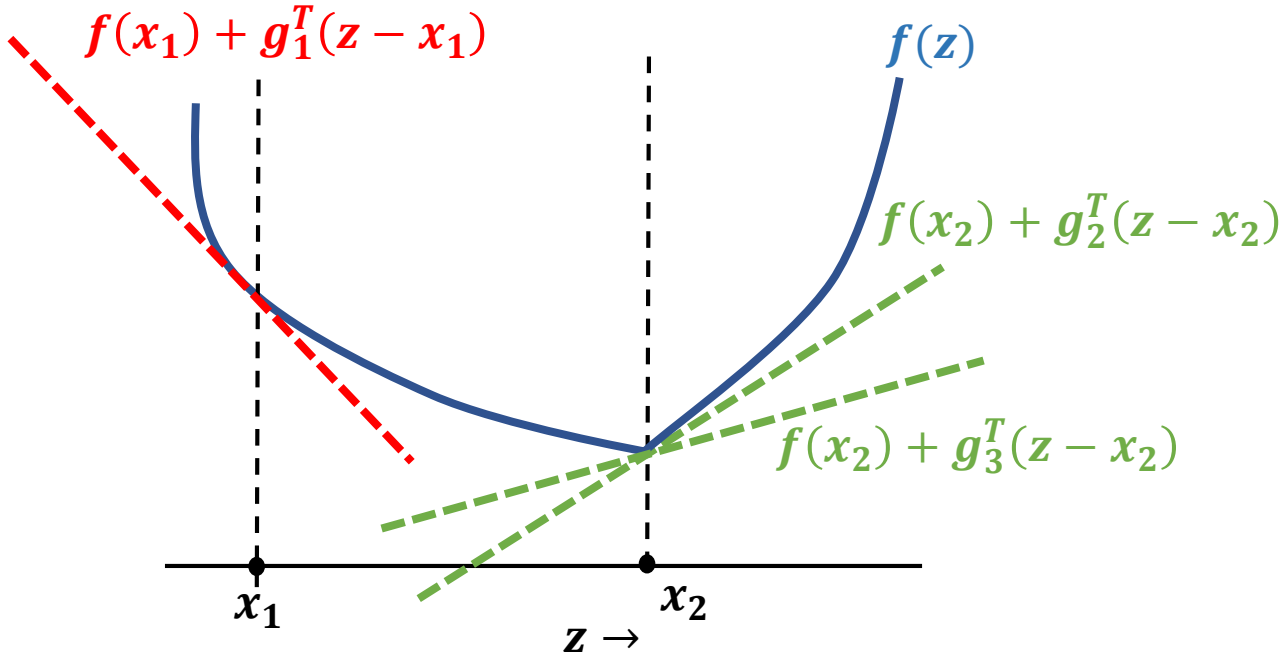
Other notations:

$$f(x) \geq f(x_0) + g^T(x_0 - x) \text{ at } x_0 \in \text{dom } f \text{ if } \forall x \in \text{dom } f$$

- The sub-gradient g is a vector that defines a slope of a supporting hyperplane (tangent line or plane) at a given point on convex function.

Sub-gradient

- At x_1, the convex function f is differentiable, and the gradient $g_1 = \nabla f(x)$ is the unique subgradient at x_1.
- At the point x_2, the function f is not differentiable. In this case, f has multiple subgradients.
  - Two subgradients, g_2 and g_3 , are examples of subgradients at x_2 , representing different slopes of supporting hyperplanes at x_2



- If f is convex and differentiable, then its gradient $\nabla f(x)$ at x is the only subgradient.
- A subgradient can exist even when f is non-differentiable at x, but the **function must be convex.**

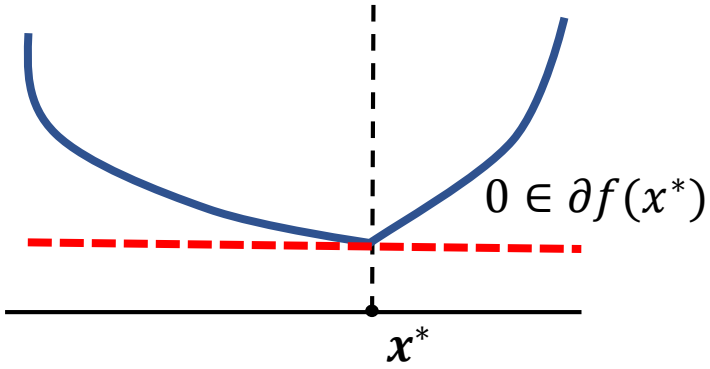
- The set of all sub-gradients at x_0 is called the **subdifferential**, denoted $\partial f(x_0)$.

$$\partial f(x_0) = \{g \in \mathbb{R}^n | f(z) \geq f(x_0) + g^T(z)(z - x_0), \forall z\}$$

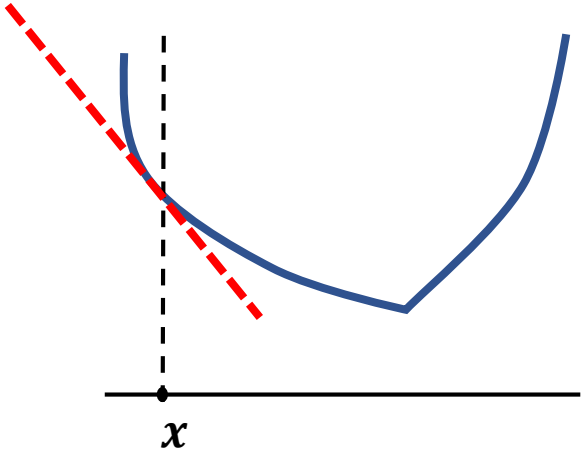
- If f is differentiable at x_0, then the gradient $\nabla f(x_0)$ is the only sub-gradient. i.e., $\partial f(x_0) = \{\nabla f(x_0)\}$
- Any point x, there can be 0, 1 or infinitely many sub-gradients.
- If f is convex and $\partial f(x) = \{g\}$ then f is differentiable at x

Sub-gradient

- **Subgradient optimal condition:**
- $0 \in \partial f(x^*)$ if and only if x^* is the global minimizer of f
  - If $0 \in \partial f(x^*)$ it means that 0 is a subgradient of f at x^*. Therefore, we have: $f(z) \geq f(x^*) + 0^T(z - x^*) \Rightarrow f(z) \geq f(x^*), \forall z \in \text{dom } f$



- If f is convex then subgradient always exists.
- If f is convex and differentiable then
  - $\partial f(x) = \{\nabla f(x)\}$
- If f is convex and $\partial f(x) = \{g\}$ then f is differentiable at x
- Any point x, there can be $0, 1$ or infinitely many subgradients.

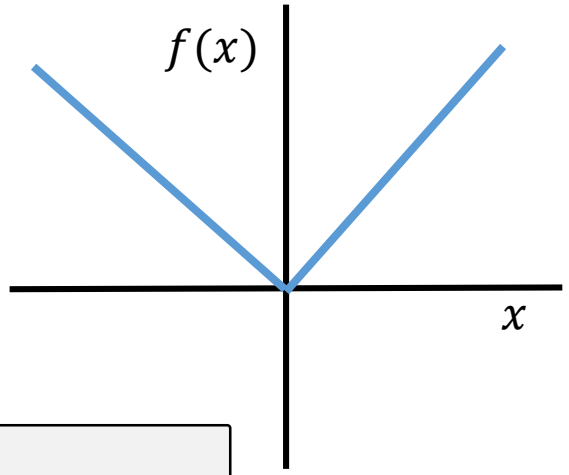


Example:

Find the subgradient of $f(x) = |x|$

- For $x < 0$
  The function $f(x) = |x| = -x$
  The gradient $\frac{d}{dx}(-x) = -1$
  Hence the subgradient g at $x < 0$ is $g = -1$
  The line $g = -1$ refers to a line with a slope of -1

- For $x > 0$
  The function $f(x) = |x| = x$
  The gradient $\frac{d}{dx}(x) = 1$
  Hence the subgradient g at $x > 0$ is $g = 1$
  The line $g = 1$ refers to a line with a slope of -1



- For $x = 0$
  - The function $f(x) = |x|$ is not differentiable at $x = 0$
  - A vector g is a subgradient of $f(x)$ at x if it satisfies the inequality:
    $f(z) \geq f(x) + g^T (z - x)$ for all points $z \in$ Dom f
  - A vector g is a subgradient at $f(x)$ at $x = 0$ if it satisfies the inequality:
    $\Rightarrow f(z) \geq f(0) + g^T (z - 0)$ for all points z
    $\Rightarrow |z| \geq 0 + g\, z$ for all points z
    $\Rightarrow |z| \geq g\, z$ for all points z
  - Since this inequality holds for $-1 \leq g \leq 1$ therefore sub gradients at $x = 0$ form the interval $[-1,1]$.

The subgradient is:

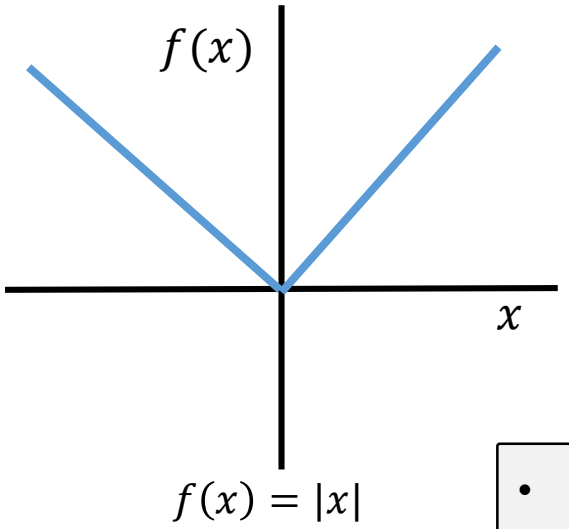
$$g = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \\ \text{Any value between } -1 \text{ to } 1, & x = 0 \end{cases}$$

- The sub-gradient g represents a slope of a supporting hyperplane to the function (convex) at that point.
- Any supporting hyperplane with slope g from $[-1,1]$ is a potential sub-gradient of $f(x)$.

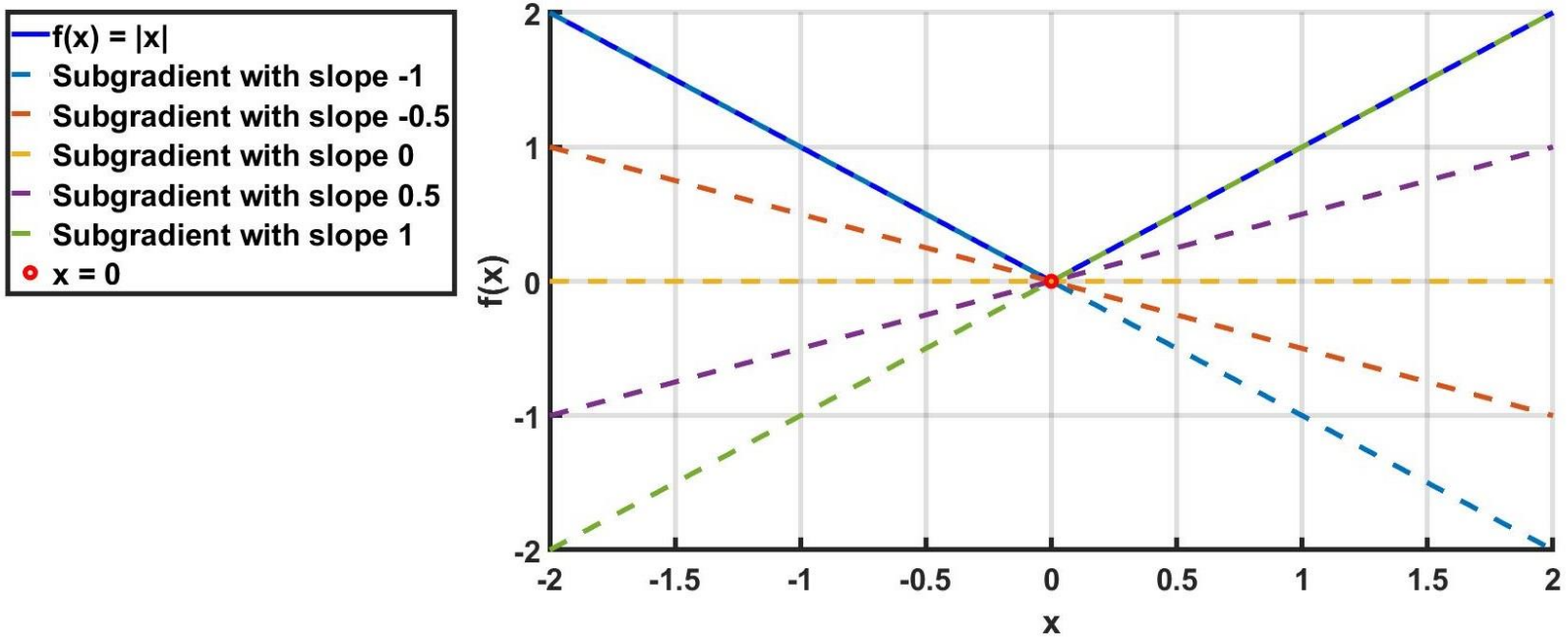
The subdifferential is:

$$\partial f(x) = \begin{cases} \{1\}, x > 0 \\ \{-1\}, x < 0 \\ [-1,1], x = 0 \end{cases}$$

Subgradients of $f(x) = |x|$ at $x = 0$



$f(x) = |x|$



Subgradients of $f(x) = |x|$ at $x = 0$

- The subgradients at $x = 0$ describe all the supporting lines that lie below or on the graph of $f(x) = |x|$ and pass through (0,0). These lines have slopes ranging from -1 to 1, ensuring they satisfy the sub-gradient condition.

- At $x_0 = 0$, let the subgradient $g = 0.5$, from the subgradient def:
- $f(x) \geq f(x_0) + g^T (x - x_0)$ for all points $x \in$ Dom f

$f(x) \geq 0 + 0.5x \Rightarrow f(x) = 0.5x$ and $f(x) > 0.5x$ both. Inequality $f(x) \geq 0 + 0.5x$ confirms that the function $f(x) = |x|$ lies above the line $y = 0.5x$. This line serves as a supporting hyperplane (supporting line in 1D) passing through $x_0 = 0$.

Example:

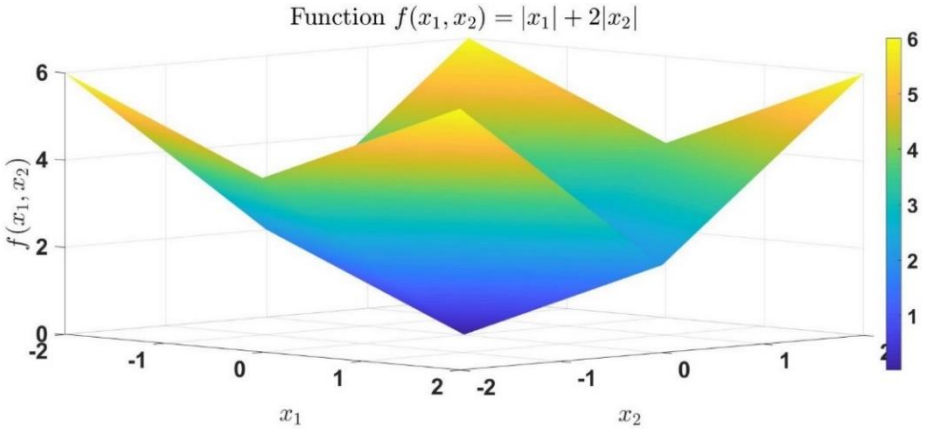
Find the sub-gradient of $f(x_1, x_2) = |x_1| + 2|x_2|$

- For x_1

$$\partial f(x) = \begin{cases} \{1\}, x_1 > 0 \\ \{-1\}, x_1 < 0 \\ [-1,1], x_1 = 0 \end{cases}$$

- For x_2

$$\partial f(x) = \begin{cases} \{2\}, x_2 > 0 \\ \{-2\}, x_2 < 0 \\ [-2,2], x_2 = 0 \end{cases}$$



The sub-gradient $g = (g_1, g_2)$ of $f(x_1, x_2)$ at any point (x_1, x_2) is

- For $x_1 > 0$ and $x_2 > 0$: $g = (1,2)$
- For $x_1 < 0$ and $x_2 > 0$: $g = (-1,2)$
- For $x_1 > 0$ and $x_2 < 0$: $g = (1, -2)$
- For $x_1 < 0$ and $x_2 < 0$: $g = (-1, -2)$
- For $x_1 = 0$ and $x_2 > 0$: $g = (g_1, 2)$ where $g_1 \in [-1,1]$
- For $x_1 > 0$ and $x_2 = 0$: $g = (1, g_2)$ where $g_2 \in [-2,2]$
- For $x_1 < 0$ and $x_2 = 0$: $g = (-1, g_2)$ where $g_2 \in [-2,2]$
- For $x_1 = 0$ and $x_2 = 0$: $g = (g_1, g_2)$ where $g_1 \in [-1,1]$ and $g_2 \in [-2,2]$

- For $x_1 > 0$ and $x_2 > 0$: $g = (1,2)$
  - The subgradient $g = (1,2)$ represents the slope of the **tangent hyperplane** at a point in the region where both $x_1 > 0$ and $x_2 > 0$.
- For $x_1 = 0$ and $x_2 = 0$: $g = (g_1, g_2)$ where $g_1 \in [-1,1]$ and $g_2 \in [-2,2]$
  - The subgradients $g = (g_1, g_2)$ *indicate a range of slopes for the* ***supporting hyperplane*** *at the non-differentiable point* $(0,0)$.

L_1 norm

L_1 norm:
For $x \in \mathbb{R}^n, f(x) = \| x \|_1 = |x_1| + |x_2| + \cdots + |x_n|$

The sub-differetial of $f(x_i) = |x_i|$ is:

$$\partial f(x_i) = \begin{cases} \{1\}, x_i > 0 \\ \{-1\}, x_i < 0 \\ [-1,1], x_i = 0 \end{cases}$$

- $\partial f(x) = \partial f(x_1) + \partial f(x_2) + \partial f(x_3) + \ldots + \partial f(x_n)$
- The sub-differential of the entire function $\partial f(x)$ is the vector formed by combining the sub-gradients of each component x_i, which results in:
- $\partial f(x) = (g_1, g_2, \ldots, g_n)$ where

$$g_i = \begin{cases} 1, x_i > 0 \\ -1, x_i < 0 \\ [-1,1], x_i = 0 \end{cases} \Rightarrow g_i = \begin{cases} \text{sign}\,(x_i), x_i \neq 0 \\ [-1,1], x_i = 0 \end{cases}$$

- Or $\partial f(x) = \{g \in \mathbb{R}^n, |g_i = 1 \text{ if } x_i > 0, g_i = -1 \text{ if } x_i < 0, g_i \in [-1,1] \text{ if } g_i = 0\}$

sign function

$$f(x) = \begin{cases} 1, x > 0 \\ -1, x < 0 \\ 0, x = 0 \end{cases}$$

- If $x_i > 0$: The sub-gradient is $+1$ for that component.
- If $x_i < 0$: : The sub-gradient is -1 for that component.
- If $x_i = 0$ The sub-gradient can be any value between -1 to 1, i.e., the sub-differential is the entire interval $[-1,1]$ for that component.

- Let $f(x_1, x_2) = |x_1| + |x_2|$
- Case 1: If $x_1 > 0$ and $x_2 > 0$ then $\partial f(x_1, x_2) = (1,1)$
  - Both components are positive, so the sub-gradient $g = (g_1, g_2) = (1,1)$
- Case 2: If $x_1 > 0$ and $x_2 < 0$ then $\partial f(x_1, x_2) = (1, -1)$
  - One component is positive, other is negative so the sub-gradient
  - $g = (g_1, g_2) = (1, -1)$
- Case 3: If $x_1 > 0$ and $x_2 = 0$ then $\partial f(x_1, x_2) = (1, [-1,1])$
  - $g = (g_1, g_2) = (1, [-1,1])$
- Case 4: If $x_1 < 0$ and $x_2 > 0$ then $\partial f(x_1, x_2) = (-1,1)$

.....

- Case: If $x_1 = x_2 = 0$
  - $g = (g_1, g_2) = ([-1,1], [-1,1])$

Sub-gradient method:

Sub-gradient method for function optimization

%Inputs

$f: \mathbb{R}^n \rightarrow \mathbb{R}$, a **non-differentiable convex function**

$x^{(0)}$: The initial guess for optimization solution

%Output: x^* a local minima of the loss function

Begin

$k \leftarrow 0$

$f_{best} \leftarrow f(x^{(0)})$

$x^* \leftarrow x^{(0)}$

While (stopping criteria k < max_it) **do**

$g^{(k)} \in \partial f(x^{(k)})$ %Compute subgradient

$x^{(k+1)} = x^{(k)} - \alpha^{(k)} g^{(k)}$ %Update next position

$f_{new} = f(x^{(k+1)})$ %Update new functional value

If ($f_{new} < f_{best}$) **then** % Apply greedy for monotonic trejectory

$f_{best} \leftarrow f_{new}$

$x^* \leftarrow x^{(k+1)}$

EndIf

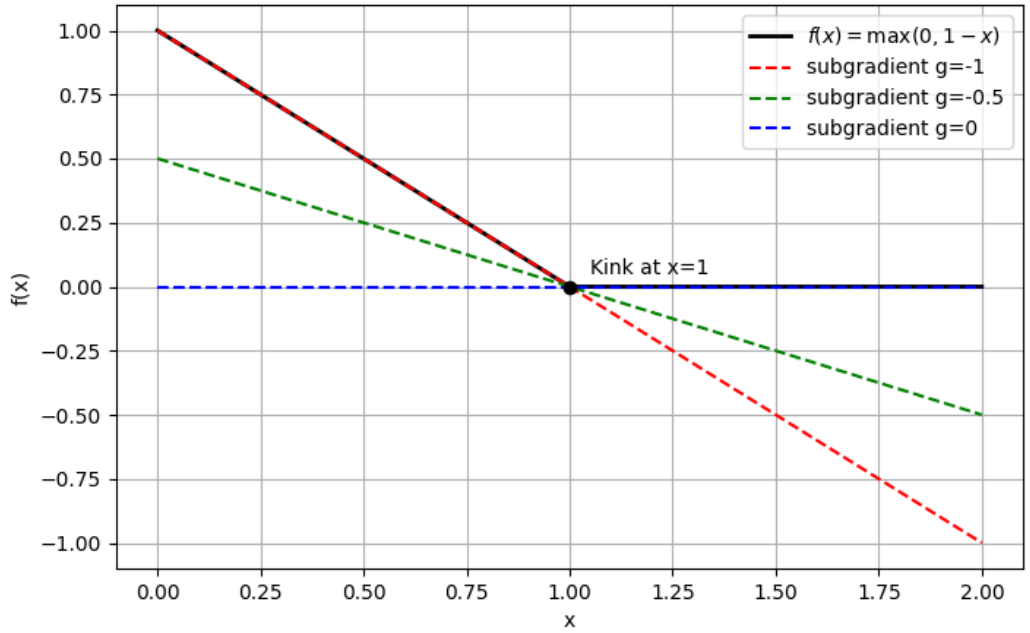
$k = k + 1;$

EndWhile

End

Hinge loss function

- $f(x) = \max(0,1-x), x \in \mathbb{R}^n$
- $f(x) = \begin{cases} 1-x, & x < 1 \\ 0, & x \geq 1 \end{cases}$
  - $g_i = \begin{cases} -1, x < 1 \\ 0, x_i > 1 \\ [-1,0], x_i = 1 \end{cases}$



- A vector g is a sub-gradient at $x = 1$ if:
- $f(z) \geq f(1) + g^T(z-1), \forall z \in Dom\ f$
- Here $f(x) = \max(0,1-x)$ so, $f(1) = \max(0,1-1) = 0$
- $f(z) \geq 0 + g^T(z-1), \forall z \in Dom$
  - $f(z) \geq g^T(z-1), \forall z \in Dom$

- If $z < 1$
- Then $f(z) = 1 - z$, therefore
- $f(z) \geq g^T(z-1) \Rightarrow (1-z) \geq g^T(z-1) \Rightarrow -(1-z) \leq -g^T(z-1)$
- $\Rightarrow (z-1) \leq -g^T(z-1) \Rightarrow 1 \leq -g^T \Rightarrow g^T \geq -1$

- If $z > 1$
- Then $f(z) = 0$, therefore
- $f(z) \geq g^T(z-1) \Rightarrow 0 \geq g^T(z-1)$
- $\Rightarrow g^T \leq 0$, since $(z-1) > 0$

$-1 \leq g^T \leq 0$, or $\partial f(1) = [-1,0]$

Lasso regression:

- Let $X^{(i)} \in \mathbb{R}$
- $\hat{Y}^{(i)} = f_W(X^{(i)}) = W_0 + W_1 X^{(i)}$
- MSE:

$$L(W) = \frac{1}{n}\sum_{i=1}^{n}\left(Y^{(i)} - \hat{Y}^{(i)}\right)^2$$

- In Lasso regression:

$$L_L(W) = \frac{1}{n}\sum_{i=1}^{n}\left(Y^{(i)} - \hat{Y}^{(i)}\right)^2 + \lambda\sum_{j=1}^{d}|W_j|$$

$$L_L(W) = \|Y - \hat{Y}\|_2^2 + \lambda\|W\|_1$$

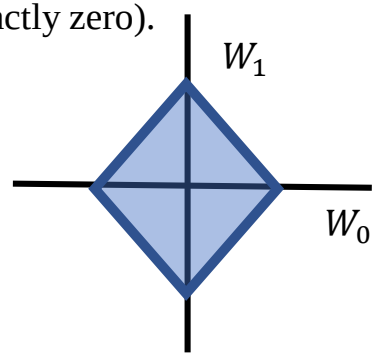
- Final Aim of parametric regression models : Find optimal set of W^* such that:

$$W^* = \arg\min_W L_L(W)$$

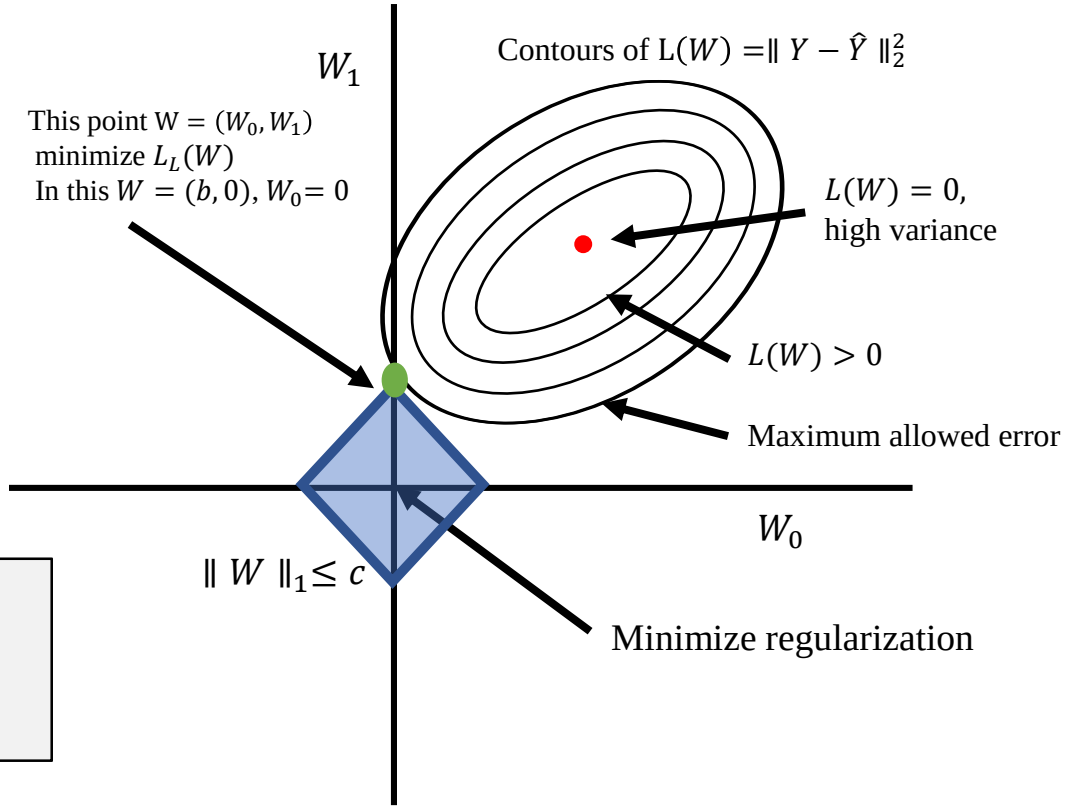
Shrink some weights exactly to zero → automatic **feature selection**

- The Lasso solution is where the contour of the MSE first touches the feasible region. Because the feasible region has corners, the contour often touches at a corner rather than along a flat edge.
- Shrink some weights exactly to zero → automatic feature selection

$\| W \|_1 \leq c$
$|W_0| + |W_1| \leq c$
The corners encourage sparsity (one of the weights can become exactly zero).



- $d = 1$: line segment $[-c, c]$
- $d = 2$: Diamond (Rhombus)
- $d = 3$: Ochtrodron
- $d = n$: $l_1$ball (cross polytope)



Sub-gradient for Lasso loss:

- l_1 regularized (**Lasso** (**least absolute shrinkage and selection operator**)) regression model:
- The regularized loss function is

$$L_{l_1}(\widetilde{W}) = \arg\min_{\widetilde{W}} \left\{ L(\widetilde{W}) + \lambda \left\| \widetilde{W} \right\|_1 \right\}$$

$$L_{l_1}(\widetilde{W}) = \arg\min_{\widetilde{W}} \left\{ \frac{1}{2} \parallel Y - \tilde{X}\widetilde{W} \parallel_2^2 + \lambda \left\| \widetilde{W} \right\|_1 \right\}$$

  - Here,
  - $\tilde{X} \in \mathbb{R}^{n \times d}$ is the design matrix (with n samples and d features),
  - $\widetilde{W} \in \mathbb{R}^d$ weight vector
  - $\left\| \widetilde{W} \right\|_1$ is L_1norm of the coefficients (promotes sparsity),
  - λ is regularization parameter controlling the trade-off between fitting the data and encouraging sparsity.

- Sub-gradient of $\left\| \widetilde{W} \right\|_1 = |w_1| + |w_2| + \cdots + |w_d|$:
- The sub gradient of $\left\| \widetilde{W} \right\|_1$ is a vector $g = (g_1, g_2, \ldots, g_d) \in \mathbb{R}^d$, where
  - each $g_i \in \{-1,1\}$ for $w_i \neq 0$
  - each $g_i \in [-1,1]$ for $w_i = 0$
- Overall, $g = \partial \left(\lambda \left\| \widetilde{W} \right\|_1 \right) = \begin{cases} \lambda, w_i > 0 \\ -\lambda, w_i < 0 \\ \lambda[-1,1],\ w_i = 0 \end{cases} \Rightarrow \begin{cases} \lambda . \text{sign}\ (w_i), \text{if}\ w_i \neq 0 \\ \lambda \times U[-1,1], if\, w_i = 0 \end{cases}$

Subgradient of $\frac{1}{2} \parallel Y - \tilde{X}\widetilde{W} \parallel_2^2$:

$$\nabla_{\widetilde{W}} \left(\frac{1}{2N} \parallel Y - \tilde{X}\widetilde{W} \parallel_2^2 \right) = -\frac{1}{N} X^T \left(Y - \tilde{X}\widetilde{W} \right) = -\frac{1}{N} X^T E$$

- Overall subgradient of Lasso loss:

$$\partial L_{l_1}(\widetilde{W}) = -\frac{1}{N} X^T \left(Y - \tilde{X}\widetilde{W} \right) + \lambda g$$

Where $g_{\text{i}} = \begin{cases} \lambda . \text{sign}\ (w_i), \text{if}\ w_i \neq 0 \\ \lambda . [-1,1], if\, w_i = 0 \end{cases}$

Sub-gradient method for Lasso loss

- Overall subgradient of Lasso loss:

$$\partial L_{l_1}(\widetilde{W}) = -\frac{1}{N} X^T (Y - \tilde{X}\widetilde{W}) + \lambda g$$

- For the optimality, the optimal weights W^* will satisfy the optimality condition:

$$0 \in \partial L_{l_1}(W^*)$$
$$-X^T(Y - \tilde{X}W^*) + \lambda g = 0$$
$$\Rightarrow X^T(Y - \tilde{X}W^*) = \lambda g$$

here $g_i \in \{1, -1\}$ for $w_i^* \neq 0$, and $g_i \in [-1,1]$ for $w_i^* = 0$.

- Apply the sub-gradient method iteratively to solve the LASSO problem using the given data.
- Let $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, Y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, regularization parameter $\lambda = 0.5$, bias $b = 0$
- Initial weights $W_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \alpha = 0.1$

- **First iteration:**
- $W^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- Compute $\nabla \left(\frac{1}{2N} \| XW^{(0)} - Y \|_2^2 \right) = \frac{1}{N} X^T (XW^{(0)} - Y) = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^T \begin{bmatrix} 0-1 \\ 0-2 \end{bmatrix} = \begin{bmatrix} -1.5 \\ -2.5 \end{bmatrix}$
- Compute $g = \partial \left(\lambda \| W^{(0)} \|_1 \right) = \begin{cases} \lambda, w_i^{(0)} > 0 \\ -\lambda, w_i^{(0)} < 0 \\ \lambda[-1,1],\ w_i^{(0)} = 0 \end{cases} \Rightarrow \begin{cases} \lambda . \text{sign}\left(w_i^{(0)}\right), \text{if } w_i^{(0)} \neq 0 \\ \lambda . U[-1,1], \text{if } w_i^{(0)} = 0 \end{cases}$
  - $g = \begin{bmatrix} \lambda[-1,1] \\ \lambda[-1,1] \end{bmatrix}$

- Position update $W^{(k+1)} = W^{(k)} - \alpha \partial L\left(W^{(k)}\right)$
- Then , $W^{(1)} = W^{(0)} - 0.1 \left(\begin{bmatrix} -1.5 \\ -2.5 \end{bmatrix} + \begin{bmatrix} \lambda \times U[-1,1] \\ \lambda \times U[-1,1] \end{bmatrix} \right)$
- Since $W^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, we can choose any value from $[-1,1]$, for each component of the sub-gradient. Let,
- First component takes 0.5 and second component take -0.7 from $[-1,1]$. Then,
- $W^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \left(\begin{bmatrix} -1.5 \\ -2.55 \end{bmatrix} + \begin{bmatrix} 0.5 \times 0.5 \\ 0.5 \times -0.7 \end{bmatrix} \right) = \begin{bmatrix} 0.125 \\ 0.285 \end{bmatrix}$

Subgradient method for Lasso loss

- **Second iteration:**
- $W^{(1)} = \begin{bmatrix} 0.125 \\ 0.285 \end{bmatrix}$
- Compute $\nabla\left(\frac{1}{2N} \parallel XW^{(1)} - Y \parallel_2^2\right) = \frac{1}{N} X^T\left(XW^{(1)} - Y\right) = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^T \left\{\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 0.125 \\ 0.285 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right\} = \frac{1}{2}\begin{bmatrix} -1.895 \\ 2.020 \end{bmatrix} = \begin{bmatrix} -0.9475 \\ 1.01 \end{bmatrix}$
- Compute $g = \partial\left(\lambda \left\| W^{(1)} \right\|_1\right) = \begin{cases} \lambda, w_i^{(1)} > 0 \\ -\lambda, w_i^{(1)} < 0 \\ \lambda[-1,1],\ w_i^{(1)} = 0 \end{cases} \Rightarrow \begin{cases} \lambda.\ \text{sign}\left(w_i^{(1)}\right), \text{if } w_i^{(1)} \neq 0 \\ \lambda.[-1,1], \text{if } w_i^{(1)} = 0 \end{cases}$
  - $g = \begin{bmatrix} \lambda \times 1 \\ \lambda \times 1 \end{bmatrix}$ since both $w_1^{(1)}$ and $w_2^{(1)}$ are positive.

- **Second iteration:**
- Position update $W^{(k+1)} = W^{(k)} - \alpha \partial L\left(W^{(k)}\right)$
- Then , $W^{(2)} = W^{(1)} - 0.1\left(\begin{bmatrix} -0.9475 \\ 1.01 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}\right)$
- $W^{(1)} = \begin{bmatrix} 0.125 \\ 0.285 \end{bmatrix} - 0.1\left(\begin{bmatrix} -0.9475 \\ 1.01 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}\right) = \begin{bmatrix} 0.08025 \\ 0.436 \end{bmatrix}$